



Density functional theory studies of the antioxidants—a review

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Received: 13 July 2021 / Accepted: 23 August 2021 / Published online: 31 August 2021
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Abstract

The following review article attempts to compare the antioxidant activity of the compounds. For this purpose, density functional theory/Becke three-parameter Lee–Yang–Parr (DFT/B3LYP) methodology was carried out instead of using pharmacological methodologies because of economic benefits and high accuracy. This methodology filtrates the compounds with the lowest antioxidant activity. At first, the Koopmans' theorem was carried out to calculate some descriptors to compare antioxidants. The energy of the highest occupied molecular orbitals (HOMO) was accepted as the best indicator, and then some studies confirmed that the highest occupied molecular orbital/lowest unoccupied molecular orbital (HOMO–LUMO) energy gap is the more precise descriptor. Although it would be better to compare spin density distribution (SDD) on the oxygen of the corresponding radical in the polarizable continuum model (PCM) to evaluate their capability to chain reaction inhibition. Next, it was mentioned that in the multi-target directed ligands (MTDLs), the antioxidant is connected to other moieties in para positions to create better antioxidants or novel hybrid compounds. Indeed, SDD was introduced as a descriptor for MTDL antioxidant effectiveness. Then, the relation between antioxidants and aromaticity was investigated. The more the aromaticity of an antioxidant, the more stable the corresponding radical is. Subsequently, in preferred antioxidant activity, it was defined that the hydrogen atom transfer (HAT) mechanism is more favored in metabolism phase I. It has been seen that the solvent model can change the antioxidant mechanism. Therefore, the solvent model is more important than the chemical structure of antioxidants, and an ideal antioxidant should be evaluated in PCM for pharmacological evaluations.

Keywords Antioxidant · Highest occupied molecular orbital · Spin density distribution · Polarizable continuum model

Introduction

Free radicals are oxygen-containing molecules with one or more numbers of unpaired electrons [1]. The uneven number allows them to change their chemical reactivity, increase their energy, and make it easily react with other molecules [2]. The oxygen-containing free radicals, including superoxide anion

radical, singlet oxygen, hydroxyl radical, and perhydroxyl radical, are called reactive oxygen species or (ROS) [3].

ROS perform their useful role in the body in their proper concentration, but the excess formation of ROS leads to serious macromolecule damages such as unwanted connections between protein and DNA and DNA breakage which causes cellular damage [4–6].

Oxidative stress is an imbalance between ROS and natural antioxidants in the body [7]. The resulting oxidative stress is believed to play a basic role in the development of aging and various degenerative diseases such as cancers and neurodegenerative diseases [8, 9].

A biological antioxidant has been defined as a molecule which is present in low concentrations compared to the oxidizable substances and transfers hydrogen or electron to neutralize ROS and delays or prevents the oxidations of the substances [10, 11]. An ideal antioxidant should quench free radical formation and chelate redux metals at physiologically relevant concentrations [12]. It should work in the aqueous or membrane domain solution [13].

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The antioxidants seem to minimize the level of the most harmful ROS on the one side while still permitting enough ROS to remain for their useful purposes. Polyphenols, carotenoids, vitamin C, and vitamin E are considered to be the main exogenous antioxidants [11].

Therefore, there is a growing interest in natural antioxidants for their clinical assessment [14]. All of the individual antioxidants, which are currently recognized, should be biologically evaluated [10]. Since all biological evaluations are time consuming, expensive, and technically demanding [15–17], a computational evaluation should be considered the easiest way for the primary assessment of natural and synthesized antioxidants.

Consequently, in this study, quantum calculations and especially the density functional theory (DFT) approach are investigated to survey antioxidant activity and the mechanism of action of different antioxidants. Thus, in this review, two strategies are applied to compare the antioxidants. The first strategy, which is known as the orbital vertical method or Koopmans' approximation method, calculates the difference between the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) energies to obtain several descriptors to compare different antioxidants [18]. The second strategy uses vertical energies, which are based on the difference of total electronic and thermal enthalpies to compare their preferred mechanism of action [18, 19].

Computational methods

Generally, density functional theory (DFT) studies constitute a family of methodologies for quantum mechanical electronic structure calculation, which can be applied to organic and main group molecules as well as more complicated systems without an additional increase in computing time [20]. It belongs to the group of so-called hybrid functional, where the high quality of the calculated results is achieved by combining DFT and the classical Hartree–Fock (HF) method. DFT methods, unlike *ab initio* methods, can be used for calculations involving metals [21].

In DFT methodology, hybrid methods consisting of the approach of the Lee, Yang, and Parr correlation functional along with Becke's three-parameter exchange functional (B3LYP) are often used for reaction calculations.

B3LYP/6-311G is a large and well-balanced triple zeta Pople-type basis set. This base provides good quality results in a reasonable time. Additionally, in order to increase the quality of the results, the standard 6-311G basis set is enhanced with polarization and diffusion functions [22]. In contrast, the function of MPW1K appears to be much better than that of B3LYP for calculations of hydrogen adsorption reactions by both PCET and HAT mechanisms [23].

In this review, the antioxidant activity of natural and synthetic antioxidants from 1996 to 2021 was evaluated theoretically. The majority of the studies calculated the electron density of the antioxidants using DFT (Table 1). In these studies, DFT can produce some meaningful descriptors to compare computational with experimental results. For example; Evangelos G. Bakalbassis et al. in their studies compared both *ab initio* and DFT methods to calculate the enthalpy of formation for *p*-hydroxycinnamic acid derivatives. Enthalpy of formation appeared to present a more considerable descriptor of antioxidant activity compared to the experimental data. They concluded that calculation at the DFT/B3LYP level of theory becomes more accurate than other models [24].

Then again, Vedernikova et al. calculated the bond lengths of antioxidant structure with the DFT/PLAP3 method. They notified that the results were in better agreement with the existing experimental data compared to the HF estimate [53]. Furthermore, Tachikawa et al. studied the Roman spectrum of the antioxidants; they concluded that the vibrational frequencies are in much better agreement with experimental spectra than those calculated based on the HF method because the HF method obtains uncertainty due to anharmonicity corrections. To correct the anharmonic effects usually scales down the theoretically calculated harmonic wavenumbers using one or more scaling factors [54].

According to Fig. 1, rosmarinic acid is an antioxidant containing hydrogen bonds in its structure. Huai Cao et al. used the DFT method to study rosmarinic acid antioxidant activity. Because of the antioxidants with hydrogen bonds in their structures, they believed that DFT methods using electron relation affect better than HF in which the electron spin is not considered [55].

It seems that DFT can achieve an excellent compromise between the computational time and the description of the electronic correlation. J.J. Fife and Olivier Holtomo et al. confirmed this assumption in separate studies [41, 43].

On the other hand, Abad Ali et al. in their study elucidated that DFT/ B3LYP method has been preferred to other methods to study antioxidant IR and NMR spectra because not only that it offers excellent chemical accuracy, but it also provides favored computational cost [56].

According to S. Anita et al., DFT/ B3LYP seems to provide accurate thermodynamical properties of the phenolic antioxidants. Furthermore, Huai Cao et al. realized that the DFT method can provide a reliable geometry and a fine electron structure to explain the structure–activity relationship (SAR) of antioxidants [57]. Moreover, Hong-Yu Zhang et al. investigated the substituent effects on phenol-binding dissociation enthalpy (BDE) [58]. This is an important thermodynamic descriptor to study antioxidant activity. Thus, they concluded that DFT can complement observations on

Table 1 Computational studies on structural and spectral properties of the antioxidant and their proper basis set from 1996 to 2021

Title	Year	Author	Basis set	References
A computational analysis of substituent effects on the O–H bond dissociation energy in phenols: polar versus radical effects	1997	Tore Brinck et al.	Ab initio (MP2-MP4) and density functional theory (DFT) method, (BLYP, B3LYP), B3LYP/6-31G**	[25]
Initiation reactions of lipid peroxidation: a Raman spectroscopic and quantum-mechanical study	1998	Hiroyasu Tachikawa et al.	Hartree-Fock (HF) method and DFT (B3LYP) method, 6-31G*	[26]
Pro oxidant toxicity of polyphenolic antioxidants to HL-60 cells: description of quantitative structure-activity relationships	1999	Sergedienne, Egle et al.	Semiempirical PM3, ab initio HF/3-21G*, and density functional theory (DFT) B3LYP/6-311G**	[27]
Local atomic and orbital reactivity indices from density functional calculations for hydrogen-bonded 1,2-dihydroxybenzene	2000	Vedernikova et al.	Kohn-Sham LCGTO-DFT program, deMon-KS3, DFT/PLAP3 method	[28]
Ab initio and density functional theory studies for the explanation of the antioxidant activity of certain phenolic acids	2001	Evangelos G. Bakalbassis et al.	HF/6-31 + G(d) and B3LYP/6-31 + G(d) levels	[29]
5-pyrimidinols: novel chain-breaking antioxidants more effective than phenols	2001	Derek A. Pratt et al.	DFT model (RO)B3LYP/6-311 + G(2d,2p)//AM1/AM1	[30]
Theoretical elucidation on the antioxidant mechanism of curcumin: a DFT study	2002	You-Min Sun et al.	DFT method, B3LYP function on the basis set of 6-31G(d,p)/B3LYP/6-311 + G (2d,2p)//AM1/AM1)	[31]
Theoretical elucidation on structure antioxidant activity relationships for indolinic hydroxylamines	2003	Hong-YuZhang et al.	B3LYP/6-31G(d,p)	[32]
Antioxidant properties of phenolic compounds: H-atom versus electron transfer mechanism	2004	Monica Leopoldini et al.	B3LYP/6-311 + +G(3df,2p)	[33]
DFT study on the antioxidant activity of rosmarinic acid	2005	Huai Cao et al.	B3LYP/6-311G*	[34]
DFT/B3LYP study of tocopherols and chromans antioxidant action energetic	2007	Erik Klein et al.	DFT method with UB3LYP	[35]
DFT study of quercetin activated forms involved in antiradical, antioxidant, and prooxidant biological processes	2007	Sebastien Fiorucci et al.	B3LYP/6-31G*	[36]
Antioxidant activity of (+)-bergenin a phytoconstituent isolated from the bark of <i>Sacoglottis uchi</i> Huber (Humireaceae)	2008	Heitor A. De Abreu et al.	B3LYP//6-311 + +G(d,p)	[37]
A density functional study of antioxidant properties on anthocyanidins	2009	Rosa Guzmán et al.	B3LYP/6-31G(d,p)	[38]
A theoretical antioxidant pharmacophore for resveratrol	2009	Auriekson N et al.	B3LYP/6-31G*	[39]
Tautomerism and radical-scavenging activity of edaravone by DFT methods	2010	Auriekson N. Queiroz et al.	B3LYP/6-31G *	[40]
Solvent effects on the antioxidant activity of 3,4-dihydroxynaphthylpyruvic acid: DFT and TD-DFT studies	2011	J.J. Fife et al.	B3LYP/6-31 + G(d)	[41]
Antioxidant activity of rooperol investigated through Cu (I and II) chelation ability and the hydrogen transfer mechanism: a DFT study	2012	Mwadhham M. Kabanda et al.	B3LYP/6-31G(d), 6-311G(d), and 6-31 + G(d,p)	[42]
DFT study of the solvent effects on the structure, UV-Vis spectra and the antioxidant activity of caffeic acid phenethyl ester and some of its derivatives	2013	Olivier Holtomo et al.	B3LYP/6-31G (d,p), (B3LYP/6-31 + G (d,p))	[43]
Antioxidant potential of orientin: a combined experimental and DFT approach	2014	R. Praveena et al.	6-311G	[44]

Table 1 (continued)

Title	Year	Author	Basis set	References
DFT calculations, spectroscopy and antioxidant activity studies on (E)-2-nitro-4-[(phenylimino)methyl]phenol	2015	Ersin Temel et al.	B3LYP/6-311G+(d,p)	[45]
A DFT investigation on the structural and antioxidant properties of new isolated interglycosidic O-(1 → 3) linkage flavonols	2016	Gabriel L. C. de Souza et al.	6-31G(d,p) 6-31 + G(d,p)	[46]
DFT/B3LYP calculations, in vitro cytotoxicity and antioxidant activities of steroidal pyrimidines and their interaction with HSA using molecular docking and multispectroscopic techniques	2017	Abad Ali et al.	(B3LYP)/6-31G(d,p)	[47]
Synthesis, spectroscopic, computational (DFT/B3LYP), AChE inhibition, and antioxidant studies of imidazole derivative	2018	Faheem Ahmad et al.	DFT/B3LYP-D3	[48]
Antioxidant properties of <i>Aloe vera</i> components: a DFT theoretical evaluation	2019	Reza Nazifi et al.	B3LYP/6-311 + + G**	[49]
DFT and QTAIM based investigation on the structure and antioxidant behavior of lichen substances atranorin, evermic acid, and diffractaic acid	2019	T.K. Shameera Ahamed	M06-2X/6-311 + + G (d,p)	[50]
Theoretical investigation on the structure and antioxidant activity of (+) catechin and (−) epicatechin – a comparative study	2020	Anitha, S et al.	B3LYP/6-311G(2d,2p)	[51]
Synthesis, antioxidant activity, and density functional theory study of some novel 4-[(benzo[d]thiazol-2-ylimino)methyl]phenol derivatives: a comparative approach for the explanation of their radical scavenging activities	2021	Asgarshamsi et al.	B3LYP/6-311 + + G**	[52]

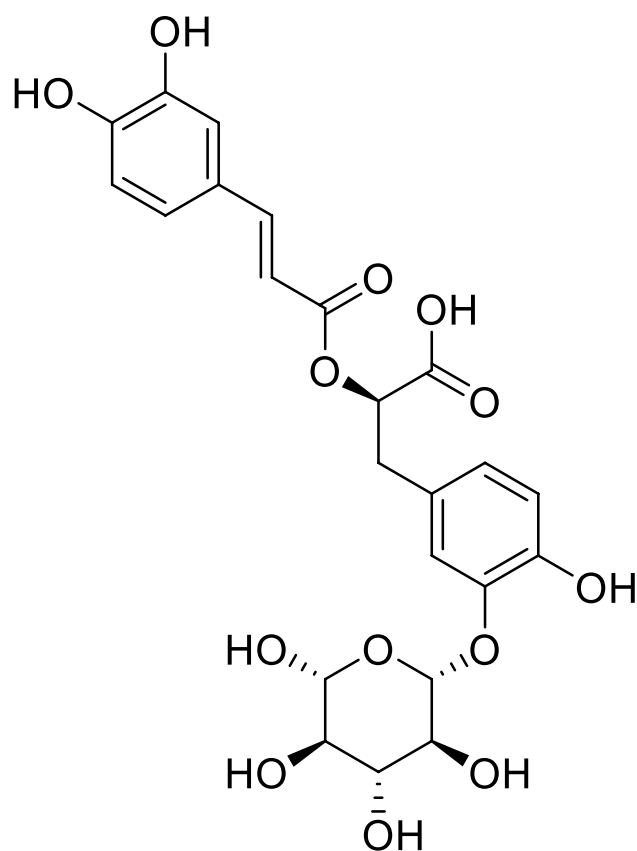


Fig. 1 The chemical structure of rosmarinic acid

the role of the characteristic groups on the structure and antioxidant activity relationship [59].

The energetics at the DFT level are much superior to Møller–Plesset perturbation theory (MP), and in most cases, they are in good agreement with high levels of theory [21].

Indeed, according to Tore Brinck et al. studies, they calculated BDE using MP2, MP4, BLYP, and B3LYP methods. They approved that the BLYP and B3LYP DFT methods provide more reliable and economical approaches for the prediction of (BDE) based on electronic and thermal enthalpies of each moiety. Indeed, despite the B3LYP method, they concluded that MP2 and MP4 methods constantly overestimate the absolute BDEs, although they provide reasonable relative BDEs. But Hong-Yu Zhang et al. in their studies demonstrated that DFT underestimates the O–H BDE to obtain the correct answer.

Finally, Faheem Ahmad et al. confirmed other consequences. They demonstrated that DFT methods along with their complementary experimental techniques play a crucial role in studying structural and spectral properties of antioxidants as well as various structures based on molecular properties of antioxidants. The hybrid functional B3LYP is most widely used for organic compounds as it offers low computational cost than post-HF methods and commonly

yields comparable results. Some of the studies on structural and spectral properties of the antioxidant and their properties are summarized in Table 1.

Frontier molecular orbital analysis of antioxidants

The energies of the highest occupied molecular orbitals (HOMO) and lowest unoccupied molecular orbitals (LUMO) of the antioxidant are important quantum chemical descriptors, which play a major role in the chemical reactivity of antioxidants. These are very significant in the study of charge transfer between antioxidants and ROS.

The first descriptor to evaluate the free radical-scavenging activity of phenolic antioxidants is HOMO energy. In a study, Fukui suggested that the most reactive site of an aromatic hydrocarbon toward electrophiles could be predicted from the spatial distribution of the electrons in its HOMO [60]. According to his approach, the site of the greatest electronic density of the HOMO would presumably correspond to the highest concentration of the most energetic electrons, and these are quite rationally expected to be the most reactive toward electrophiles and radicals [61]. Nazifi et al. and Akhtari et al. in separate research compared the HOMO energy of different antioxidants to characterize the ability of electron-giving of antioxidants to compare their free radical-scavenging efficiency [49, 62].

For reactions with nucleophiles, the studies have been concentrated on the lowest unoccupied molecular orbital (LUMO) of the substrate. HOMOs and LUMOs are characterized as “frontier molecular orbitals” (FMO) [63]. Thus, other studies have used the frontier molecular orbital (FMO) energies to indicate which site of scavenging in the antioxidant molecule is more active qualitatively. HuaiCao et al. indicated that the lower the HOMO is, the weaker the electron-donating ability of the site of scavenging is [55]. On the contrary, the higher HOMO implies that the site of scavenging is a promising electron donor. They confirmed that LUMO presents the ability of a molecule to receive electrons [64].

According to the FMO theory, the formation of the transition state in chemical reactions is due to the interaction between the HOMO and LUMO orbitals between antioxidants and free radicals. Thus, it seems that the energy difference between the HOMO and LUMO should be another descriptor for anti-oxidative activity. This difference is termed the HOMO–LUMO gap. According to Zainuri et al., by using the HOMO–LUMO gap, one can easily determine the excitation energy of an antioxidant at its ground state [65]. The HOMO–LUMO gap of different compounds can be used to predict and compare the strength and stability of antioxidants. Indeed, according to Fazlul Hug’s study,

a higher HOMO–LUMO gap indicates that the antioxidant would be more kinetically inert than other antioxidants [66]. While antioxidants with small HOMO–LUMO gaps are generally reactive according to Zainuri et al.

In the Hartree–Fock theory, the energy needed to remove an electron from a HOMO in an antioxidant is obtained by inserting the appropriate Hartree–Fock energy. The same procedure is for the gain of an electron [63]. Therefore, two energetic descriptors are defined as ionization potential (IP) and electron affinity (EA). According to Koopmans' theorem, the negative amount of energy of HOMO and LUMO signifies the ionization energy and electron affinity, respectively [67].

IP can be inferred as having less tendency to lose electrons. In other words, an electron removes from the HOMO of the neutral form of antioxidant, and this molecule changes to the radical cation form. The lower IP, the stronger affinity to change to radical cation form.

The electron affinity values are another descriptor expressed as the amount of energy released or spent when an electron is abstracted [68]. The greater EA of an antioxidant means it abstracts electrons more easily than other molecules. The electron is absorbed from a free radical to the LUMO of the neutral form of antioxidant. According to Koopmans' theorem, IP and EA were used to calculate another important descriptor, i.e., the hardness (η) and the softness (S). Hardness means the resistance of the antioxidant structure to the charge transfer while softness indicates affinity to charge transfer to free radical [69].

Nazifi et al. on their study confirmed that hard antioxidant has a large HOMO–LUMO gaps, and thus show a low tendency to charge transfer and act as weak antioxidants. By contrast, the soft molecules have lower HOMO–LUMO gaps and can easily transfer electrons to free radicals, indicating higher scavenging ability [49]. Table 2 summarizes different natural and synthetic antioxidant descriptors to compare their antioxidant activity. Resveratrol and quercetin were evaluated in separated antioxidant activity.

The FMO theory has been applied extensively to describe a variety of antioxidant reactions. However, some studies pointed out some weaknesses of this theory. For instance, Dewar presented a very critical analysis and concluded that the FMO theory does not provide a satisfactory explanation of chemical reactivity [63, 83].

A HOMO is composed of contributions from the atoms that comprise a molecule and typically extends over a large portion of the molecule [84]. Although it had always been assumed that HOMO is generally the simplest indication for the most “energetic” electrons, Bulat et al. elucidated that the HOMO is not a reliable parameter to locate the most energetic electrons of antioxidants; the most energetic electrons of an antioxidant may be in other occupied and unoccupied molecular orbitals in that molecule [63, 85].

According to this study, the Hartree–Fock theory assumes that the HOMO of antioxidants is not affected by the loss of an electron. In fact, Koopmans' theorem elucidated that in removing an electron in an antioxidant, there is just HOMO of the antioxidant itself and ignores other occupied orbitals. On the contrary, the theory neglected the relaxation of the negative ion that is produced and the increase in the energy resulting from the production of this ion relative to the neutral molecule [63].

Finally, Bulat et al. proposed to use average local ionization energy to investigate the regioselectivity and relative reactivity of the antioxidants. This reactivity descriptor was introduced first by Sjoberg et al. in 1990 [86]. It is the average energy that is needed to remove an electron at the point in the space of an atom or molecule. This reactivity descriptor focused on the point in space rather than a particular atomic or molecular orbital. Indeed, the contributions of all of the atom's or molecule's electrons are considered [87, 88]. Furthermore, it explicitly takes account of all of the molecular orbitals and their energies, rather than just the HOMO [63].

Spin density distribution

The promised antioxidant should be able to stop the chain reactions, which can cause many diseases. During the free radical scavenging, the antioxidant itself becomes a radical; but it does not start chain reactions due to its stability [89].

Spin density distribution (SDD) is a descriptor, which shows the stability of antioxidants after hydrogen atom abstraction with the free radicals. SDD can provide information about the speed of radical-scavenging reactions. More delocalized SDD results in easier radical formation and faster speed of scavenging reaction [90, 91].

Subsequently, the studies also show that SDD can provide information about the reactivity of various scavenging sites in the antioxidant molecule with different hydroxyl groups (polyphenols) [92].

In the study performed in 2016, Saqib et al. calculated the SDD on the O atom of various free radicals of chlorogenic acid. This study indicates the direct relation between bonding dissociation enthalpy and SDD on oxygen after the abstraction of the H atom [93].

It was indicated that the lowest SDD on atoms of antioxidant could be due to the hyperconjugation between the aromatic rings or double bonds. Hyperconjugation delocalized the SDD from oxygen atom to other atoms of the structure and increase the stability of radicals. Moreover, the comparison shows higher delocalization of the SDD agrees with the easier formation of the radical.

Table 2 The antioxidant descriptor of natural and synthetic antioxidants to compare their efficacy. The energy values are converted to electron volts (eV)

Molecules	E _{HOMO}	E _{LUMO}	E _g	IP	Ea	S	H	Basis set	Reference
Resveratrol	−5.48	−1.45	4.03	5.48	1.45	0.24	2.01	B3LYP/6-31G**	[57]
4'-radical of resveratrol	−5.64	−1.86	3.78	5.64	1.86	0.26	1.89	B3LYP/6-31G**	[57]
5-radical of resveratrol	−5.89	−1.83	4.06	5.89	1.83	0.24	2.03	B3LYP/6-31G**	[57]
Rosmarinic acid	−5.57	−1.79	3.78	5.57	1.79	0.26	1.89	B3LYP/6-31G*	[55]
Rosmarinic acid 2-radical	−5.74	−2.09	3.31	5.74	2.09	0.27	1.82	B3LYP/6-31G*	[55]
Rosmarinic acid 1-radical	−5.71	−1.76	3.95	5.71	1.76	0.25	1.97	B3LYP/6-31G*	[55]
Rosmarinic acid 4'-radical	−5.65	−1.76	3.89	5.65	1.76	0.25	1.94	B3LYP/6-31G*	[55]
Catechin	−5.63	−0.38	5.25	5.63	0.38	0.19	2.62	PBEg/CBSB2**//PBEg/CBSB4	[70]
Epicatechin	−5.69	−0.44	5.25	5.69	0.44	0.19	2.62	PBEg/CBSB2**//PBEg/CBSB4	[70]
Luteolin	−5.98	−1.90	4.08	5.98	1.90	0.24	2.04	B3LYP/6-31 + + G**	[71]
Quercetin	−5.71	−3.17	2.54	5.71	3.17	0.39	1.27	B3LYP/6-31 + + G**	[72]
Taxifolin	−5.98	−1.90	4.08	5.98	1.90	0.24	2.04	B3LYP/6-31 + + G**	[72]
Resveratrol	−5.21	−1.15	4.06	5.21	1.15	0.24	2.03	B3LYP/6-31G*	[39]
Mearnsetin	−5.80	−2.1	3.7	5.80	2.1	0.27	1.85	B3LYP/6-311G	[73]
Myricetin	−5.70	−2.1	3.6	5.70	2.1	0.27	1.80	B3LYP/6-311G	[73]
Quercetin	−5.70	−2.3	3.4	5.70	2.3	0.29	1.7	B3LYP/6-311G	[74]
Apigenin	−6.15	−1.98	4.17	6.15	1.98	0.24	2.08	B3LYP/6-311G(d,p)	[19]
Scutellarein	−5.93	−1.98	3.15	5.93	1.98	0.25	1.97	B3LYP/6-311G(d,p)	[19]
Chrysoeriol	−6.02	−1.95	4.07	6.02	1.95	0.24	2.03	B3LYP/6-311G(d,p)	[75]
Hispidulin	−6.07	−1.97	4.1	6.07	1.97	0.24	2.05	B3LYP/6-311G(d,p)	[75]
Homogentisic acid	−5.69	−1.1	4.58	5.69	1.1	0.21	2.29	B3LYP/6-311G	[76]
Crocin	−5.26	−2.84	2.42	5.26	2.84	0.41	1.21	B3LYP/6-31G(d,p)	[62]
Mearnsetin	−5.83	−2.1	3.73	5.83	2.1	0.26	1.86	B3LYP/6-31G(d,p)	[62]
Crenulatin-a	−5.37	−1.77	4.60	5.37	1.77	0.27	1.8	B3LYP/6-31G(d)	[62]
Crenulatin-b	−5.26	−1.72	4.54	5.26	1.72	0.28	1.77	B3LYP/6-31G(d)	[62]
Orientin	−6.29	−2.17	4.12	6.29	2.17	0.24	2.06	B3LYP/6-311G(d,p)	[62]
Vitexin	−6.38	−2.14	4.24	6.38	2.14	0.23	2.12	B3LYP/6-311G(d,p)	[62]
(E)-2-nitro-4-phenylimino methyl phenol	−6.52	−3.30	3.22	6.52	3.30	0.31	1.61	B3LYP/6-311G + (d)	[77]
4-(benzo[d]thiazol-2-yl)-7-hydroxy-2H-chromen-2-one	−5.279	−3.983	1.296	5.27	3.98	0.78	0.64	B3LYP/3-21G	[78]
4-hydroxy-benzoic acid (4-diethylamino-2-hydroxy-benzylidene)hydrazide	−5.02	−1.22	3.891	5.02	1.22	0.26	1.9	B3LYP/6-311G**	[79]
(E)-1-(4-bromophenyl)-3-(4-iodophenyl)prop-2-en-1-one	−9.334	−6.776	2.558	9.33	6.77	0.39	1.28	B3LYP/631Gp(d,p)/Lanl2dz(f)	[80]
Aloesone	−6.67	−1.06	5.07	6.67	1.06	0.17	2.80	B3LYP/6-311 + + G(d,p)	[49]
Aloeemodin	−6.6	−3.11	3.48	6.6	3.11	0.28	1.74	B3LYP/6-311 + + G(d,p)	[49]
Isoeuletherol	−5.97	−1.88	4.09	5.97	1.88	0.24	2.04	B3LYP/6-311 + + G(d,p)	[49]
((1-(4-Methoxyphenethyl) piperidin-4-ylamino)methyl)-5-hydroxy-1-methylpyridin-4(1H)-one	−6.020	−1.142	4.877	6.02	1.142	0.20	2.43	B3LYP/6-311 + + G(d,p)	[81]
5-(hydroxy)-2-((1-isopropylpiperidin-4-ylamino)methyl)-1-methylpyridin-4(1H)-one	−5.998	−1.141	4.857	5.99	1.141	0.20	2.42	B3LYP/6-311 + + G(d,p)	[81]
5-hydroxy-1-methyl-2-((phenethylamino)methyl)pyridin-4(1H)-one	−6.144	−1.149	4.994	6.14	1.149	0.20	2.49	B3LYP/6-311 + + G(d,p)	[81]
((4Hydroxyphenethylamino)methyl)-5-hydroxy-1-methylpyridin-4(1H)-one	−6.143	−1.172	4.971	6.14	1.172	0.20	2.48	B3LYP/6-311 + + G(d,p)	[81]
((Tricyclo[3.3.1.1.3,7] decane-3-ylamino)methyl)-5-hydroxy-1-methylpyridin-4(1H)-one	−6.129	−1.069	5.060	6.12	1.069	0.19	2.52	B3LYP/6-311 + + G(d,p)	[81]
Kojic acid	−6.870	−1.750	5.120	6.87	1.750	0.19	2.56	B3LYP/6-311 + + G(d,p)	[81]
2-((4 Hydroxyphenethylamino)methyl)-5-hydroxy-1-methylpyridin-4(1H)-one	−6.143	−1.148	4.994	6.14	1.148	0.20	2.49	B3LYP/6-311 + + G(d,p)	[81]

Table 2 (continued)

Molecules	E _{HOMO}	E _{LUMO}	E _g	IP	Ea	S	H	Basis set	Reference
4-[(Benzo[d]thiazol-2-ylimino) methyl] phenol	−6.32	−2.56	3.76	6.32	2.56	0.26	1.88	B3LYP/6–311 + + G(d,p)	[82]
4-[[((6-Fluorobenzo[d]thiazol-2-yl) imino) methyl]phenol	6.34	−2.24	3.75	6.34	2.24	0.24	2.05	B3LYP/6–311 + + G(d,p)	[82]
4-[[((6-Methoxybenzo[d]thiazol-2-yl) imino) methyl]phenol	−5.95	−2.22	3.73	5.95	2.22	0.26	1.86	B3LYP/6–311 + + G(d,p)	[82]
4-[[((6-Ethoxybenzo[d]thiazol-2-yl) imino) methyl]phenol	−5.95	−2.29	3.66	5.95	2.29	0.26	1.86	B3LYP/6–311 + + G(d,p)	[82]
4-[[((6-(Trifluoromethoxy)benzo[d]thiazol-2-yl) imino)methyl]phenol	−6.50	−2.25	4.25	6.5	2.25	0.23	2.12	B3LYP/6–311 + + G(d,p)	[82]

On the other hand, SDD can give an idea of promised design of hybrid compounds. Antioxidants can be used to connect to another pharmacophore to design novel multi-target directed ligands. In these compounds, higher delocalization of SDD is a sign for designing promising compounds.

It has been reported that the para-substituted polyphenol compounds were exciting candidates for the evaluation of antioxidant potentials. Moreover, bridging the phenolic group in the para position can increase the antioxidant activity of phenols due to the more efficient overlap of the substituent orbital within the aromatic system [94]. Thus, the connection of a hyperconjugated system including rings and double bonds with para positions of polyphenols should be efficient. In 2021, Asgarshamsi et al. synthesized some novel 4-[(benzo[d]thiazol-2 ylimino)methyl]phenol derivatives. In these structures, SDD is mostly distributed to the O atom in the para position, double bond, and B-benzene ring [82]. Indeed, Fassihi et al. in 2020 designed new antioxidants based on 5-hydroxypyridine-4-one moiety as the core connected to other moieties in the para position of the core. Their SDD demonstrated their efficient drug design [95]. Figure 2 shows their rational design [96].

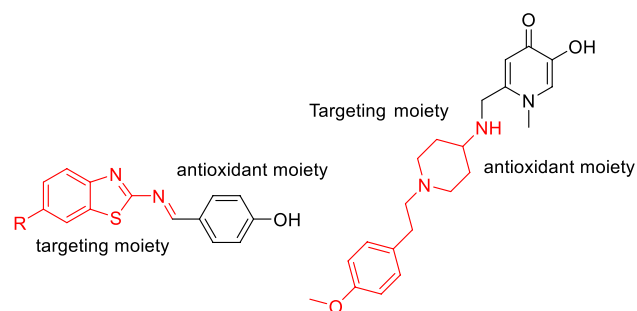


Fig. 2 Rational design of multi-target drug ligands (MTDLs) using antioxidant moiety

Finally, the amounts of the SDD on the radical species of different antioxidants are compared in Table 3.

Aromaticity and antioxidant activity

Olennikov et al. found that compounds with a high degree of aromaticity exhibited noticeable antioxidant activity [98].

Thus, the aromaticity of each ring should have calculated using the following descriptors such as the harmonic oscillator model of aromaticity (HOMA), aromatic fluctuation index (FLU), and para-delocalization index (PDI) [99, 100].

The closer the HOMA value to 1 indicates the more aromatic the ring is. Moreover, the higher the value of the PDI, the delocalization will be larger and the stronger the aromaticity is. Furthermore, lower FLU corresponds to stronger aromaticity.

Few studies have been carried out about the relation between antioxidant activity and aromaticity of the rings. Through them, Zoran Markovic et al. designed a quantitative structure–activity relationship (QSAR) model to describe the relationship between antioxidant activity with thermodynamic and aromaticity descriptors [101].

Zborowsky et al. investigated the aromaticity of hydroxyl pyridine-4-one; they concluded that tautomerism in hydroxypyridinone ring leads to enough level of aromaticity resulting in a catechol-like behavior that provides them with good chelating and radical-scavenging properties. Finally, Fassihi et al. synthesized their proposed antioxidants in 2020 [95, 102].

Antioxidant mechanisms

Fifen et al. recognized that two main mechanisms are involved in the reactive species-scavenging processes; the hydrogen atom transfer (HAT) processes and the proton

Table 3 The amount of spin density distribution on the radical species of different antioxidants

Compounds	BDE kcal/mol	Spin density	Basis set	References
1OH-radical chlorogenic acid	68.33	0.378	B3LYP/6-311G	[93]
2OH-radical chlorogenic acid	87.66	0.511	B3LYP/6-311G	[93]
3'OH-radical chlorogenic acid	105.35	0.945	B3LYP/6-311G	[93]
4'OH-radical chlorogenic acid	97.51	0.889	B3LYP/6-311G	[93]
Aloesone	398.61	0.63	B3LYP/6-311 + + G**	[91]
Aloe-emodin R1	343.95	0.67	B3LYP/6-311 + + G**	[91]
Aloe-emodin R2	399.64	0.78	B3LYP/6-311 + + G**	[91]
Isoeuletherol	395.79	0.55	B3LYP/6-311 + + G**	[91]
Caffeic acid radical 3	83.19	0.412	(U)B3LYP, 6-31G(d,p)	[97]
Caffeic acid radical 2	71.12	0.310	(U)B3LYP, 6-31G(d,p)	[97]
3,4 dihydroxybenzoic acid radical 2	74.44	0.337	(U)B3LYP, 6-31G(d,p)	[97]
3,4 dihydroxybenzoic acid radical 3	83.67	0.428	(U)B3LYP, 6-31G(d,p)	[97]
Ferulic acid radical	76.58	0.322	(U)B3LYP, 6-31G(d,p)	[97]
Vanillic acid radical	79.41	0.362	(U)B3LYP, 6-31G(d,p)	[97]
Sinapic acid radical	76.87	0.302	(U)B3LYP, 6-31G(d,p)	[97]
Syringic acid radical	79.42	0.341	(U)B3LYP, 6-31G(d,p)	[97]
Para coumaric acid	81.20	0.360	(U)B3LYP, 6-31G(d,p)	[97]
Para-hydroxybenzoic acid radical	85.12	0.429	(U)B3LYP, 6-31G(d,p)	[97]
Vitamin C radical 2	69.94	0.277	(U)B3LYP, 6-31G(d,p)	[97]
Vitamin E radical	76.91	0.401	(U)B3LYP, 6-31G(d,p)	[97]
Vitamin C radical 1	80.53	0.405	(U)B3LYP, 6-31G(d,p)	[97]
Vitamin B6 radical 2	96.58	0.887	(U)B3LYP, 6-31G(d,p)	[97]
Vitamin B6 radical 1	96.65	0.883	(U)B3LYP, 6-31G(d,p)	[97]
Vitamin B6 radical 3	97.15	0.639	(U)B3LYP, 6-31G(d,p)	[97]
Vitamin C radical 4	98.22	0.875	(U)B3LYP, 6-31G(d,p)	[97]
Vitamin C radical 3	103.20	0.889	(U)B3LYP, 6-31G(d,p)	[97]
Vitamin B3 radical	104.03	0.552	(U)B3LYP, 6-31G(d,p)	[97]

transfer (PT) process. Approximately, the hydrogen atom transfer processes can be divided into a homolytic hydrogen atom transfer (HHAT) and a heterolytic hydrogen atom transfer. The latter can either be a single-electron transfer followed by a proton transfer (SET-PT), or a sequential proton loss electron transfer (SPLET). The proposed antioxidant mechanisms are depicted below through the following equations (Fig. 3) [103]:

According to Fig. 3, in all these reactions, the antioxidant is converted to a neutral free radical, which is more stable and less reactive than other reactive radical species (*R*). To study the antioxidant mechanisms of compounds, some thermodynamic descriptors are calculated which will be described below in detail. These reactions consist of two

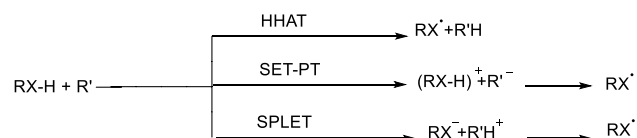
steps; thus, the descriptors which show the first step of the mechanism are rate-limiting and determine the preferred reaction pathway (IP, PA). SET-PT and SPLET mechanisms are solvent-dependent and may coexist in some chemical or biological systems.

Finally, according to Sadasivam et al. studies, it was notified that the formation of the radical species $RX^\bullet$ seems to take place in all three mechanisms (HAT, SPLET, ET-PT). Therefore, regardless of the antioxidant mechanisms, the radical-scavenging activity can also be dependent on the stability of the $RX^\bullet$ species. To evaluate this phenomenon, the SDD of the unpaired electron in the $RX^\bullet$ radical should be analyzed [19].

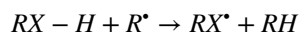
HAT mechanism

Hydrogen atom transfer (HAT) refers to any reaction in which a substrate can lead to the abstraction of free hydrogen radicals.

One of the most important examples of HAT reactions is oxidative reactions involving cytochrome P450 containing an iron(V)-oxo unit which plays a key role in drug

**Fig. 3** The depiction of preferred antioxidant mechanism

metabolism. For hydroxylation reaction in phase I metabolism, at first, carbon-hydrogen bond (C–H) must be homolytically cleaved by the cytochrome P450 enzyme, which then the neutral compound (*R*) is converted to a free radical (*R*[•]). The first mechanism involves HAT from the hydroxyl or the C–H group.



In this reaction, *RX*–H represents an antioxidant, *X* represents O, S, N, or C, and *R* represents a reactive species (RS). OH, SH, or NH groups are constantly available for oxidizing agents, but sometimes, these antioxidants can be converted to their tautomer. For example, edaravone (3-methyl-1-phenyl-2-pyrazolin-5-one) seems to be in amine and keto form. (Fig. 4).

BDE is a numerical parameter related to the HAT mechanism which is calculated using the electronic and thermal enthalpies summation as follows:

$$BDE = H_r + H_h - H_n$$

where *H_r* is the enthalpy of the radicals generated through H-atom abstraction, *H_h* is the enthalpy of the hydrogen atom; *H_n* is the enthalpy of the related neutral compound. Indeed, the total electronic enthalpy of the H atom was considered 1312.7 kJ mol^{−1}.

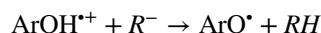
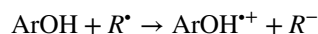
Low intramolecular hydrogen bonding in the antioxidant structure can help to dissociate the O–H bond thus decreases the BDE value. According to the study carried out by Houssein Boulebd et al., the implicitly of polar and nonpolar solvents affected the antioxidant mechanism. It was found that in the gas and benzene phases, all the studied compounds preferred to undergo the HAT mechanism [104].

It can be mentioned that in drug metabolism, the C–H with lower BDE will show more probable sites of metabolism than the other C–H present in the substrate. The steric factor plays an important role in the location of the main hydrogen uptake site based on access to individual C–H sites. It is unlikely that the major hydrogen abstraction will

occur at a C–H site with a low BDE if it is situated at the center of a bulky substituent [105].

SET-PT mechanism

Electron and then proton transfer from the phenolic hydroxyl oxygen (SET-PT) is the other prominent antioxidant mechanism for this class of antioxidant scaffolds:



Numerical parameters related to the SET-PT mechanism are ionization potential (IP) and proton dissociation enthalpy (PDE) are calculated using electronic and thermal enthalpy of related compounds as follows:

$$IP = H_n - H_{rad.cat}$$

$$PDE = H_r + H_{pro} - H_{rad.cat}$$

where *H_r* is the enthalpy of radical, *H_{rad. cat}* is the enthalpy of radical cation, and *H_{pro}* is the proton enthalpy, which is − 1090 kJ mol^{−1}.

Moreover, in polar solvents, intermolecular hydrogen bonds could be formed between the antioxidant and a solvent molecule, which will hamper the H-atom transfer process and consequently facilitates the SET-PT [106].

SPLET mechanism

Numerical parameters in connection with the SPLET mechanism are electron transfer enthalpy (ETE) and proton affinity (PA) are calculated using electronic and thermal enthalpy of related compounds as follows:

$$PA = H_{an} + H_{pro} - H_n$$

$$ETE = H_r + H_e - H_{an}$$

In this equation, *H_{an}* is the enthalpy of the anion and *H_e* is the enthalpy of the electron, which is − 236 kJ mol^{−1}.

This procedure is solvent-dependent because of the charged species formation. Obviously, solvation, *pK_a*, and formation possibility of charged species have major effects on this mechanism [96]. Erik Klein et al. studied the antioxidant activity of tocopherols. For α-tocopherol, the reaction enthalpies in water were computed and compared to the gas phase; they concluded that water causes severe changes in the enthalpies of studied compounds [35]. From the thermodynamic point of view, the SPLET mechanism represents the most probable mechanism in PCM water.

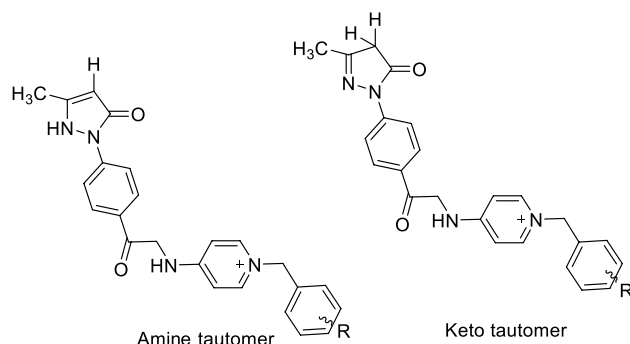


Fig. 4 Keto and amine form of edaravone

Table 4 Different antioxidants and their mechanisms of action

Compound	Mechanism	Model	References
Aloe vera component	SPLET	PCM	[91]
Amlodipine besylate	HAT	Gas	[107]
Tocopherol	SPLET	PCM/Gas	[108]
Rosmarinic acid	HAT	Gas	[34]
Orientin	HAT	Gas	[109]
Rooperol	HAT	Gas	[110]
4-[(benzo[d]thiazol-2-ylimino) methyl]phenol derivatives	SPLET	PCM	[82]
Moracin T	SPLET	Gas	[111]
Hydroxypyridine-4-one derivatives	SPLET	PCM	[95]

In sum, as mentioned above, it seems that the mechanism of antioxidant activity seriously depends on the medium. Antioxidants can react via different mechanisms in different media. Polar solvents such as the polarizable continuum model (PCM) stabilize ionization of the 3-OH with the consequent electron transfer reaction from this site (SPLET) [91].

Indeed, it should be noted that the PCM presents good accuracy, reliability, adaptability, and a reduced computational effort in describing solvent effects. Furthermore, the PCM method has been widely adopted in recent years, especially in the description of the thermodynamic characteristics of solvation [106]. Table 4 shows some natural and synthetic antioxidants and their mechanism.

Conclusions

This review article attempts to compare the antioxidant activity of the compounds. As it was mentioned in [Introduction](#) section, instead of pharmacological methodologies, it is better to use a simulation study using DFT to compare the antioxidant activity of the compounds. This methodology filters out compounds with the least antioxidant activity so that no further pharmacological experiments are required. It was recognized that DFT/B3LYP has been used for the majority of studies because of economic benefits and high accuracy. Subsequently, the Koopmans' theorem was used to calculate some descriptors to compare the antioxidant efficacy of the natural and synthetic compounds. At first, the energy of HOMO was accepted as the best indicator for choosing antioxidants. Then, some studies confirmed that between HOMO–LUMO energy gap is the more precise descriptor. It seems that researchers should calculate the average local ionization energies to compare the antioxidant activity because this descriptor has the best predictive power than FMO descriptors.

When an antioxidant turns into the radical moiety via hydrogen abstraction, it should be kinetically inert; it should

be evaluated not to start chain reactions. Although, it would be better to compare SDD on the oxygen of the corresponding radical in PCM.

Then, it was mentioned that in MTDLs, the antioxidant is connected to other moieties in para positions to create better antioxidants or novel hybrid compounds with pharmacological activity. Indeed, SDD was introduced as a descriptor for MTDL antioxidant effectiveness.

Another neglected point in the DFT calculation is the aromaticity of the antioxidants. A few articles have been published about the relation between antioxidant and aromaticity, mentioning that the more the aromaticity of an antioxidant, the more stable the corresponding radical is. However, it seems that more research should be done in this regard in the future.

Overall, it was defined that the HAT mechanism is more favored in metabolism phase I. It has been further seen that the solvent model can change the preferred antioxidant mechanism. To determine the antioxidant mechanism, the solvent model is more important than the chemical structure of antioxidants. Furthermore, for pharmacological evaluation, an ideal antioxidant should be evaluated in PCM.

Author contribution The study's conception, design, data collection, analysis, writing, editing, review, and supervision were performed by Mohammad Hossein Asgarshamsi. The data collection, analysis, writing, editing, and review were performed by Samira Mahmoudi. The analysis, review, and editing were performed by Mehrdad Mohammadpour Dehkordi.

Availability of data and material Not applicable.

Code availability Not applicable.

Declarations

Ethics approval Not applicable.

Consent to participate All authors agreed to contribute to the work, critically drafted the manuscript, and revised it.

Consent for publication All authors approved this version to be published.

Conflict of interest The authors declare no competing interests.

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